

The prompt is the protocol: a disclosure standard for clinician-investigators using agentic large language models

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Abstract. A single clinician-investigator can in 2026 assemble an end-to-end agentic-LLM research workflow on commodity tooling; current biomedical-journal AI policies classify most of what such a workflow does as prohibited, and we propose a Disclosure 2.0 standard — prompts, model identifier, commit history, reviewer-subagent transcripts, audit log, tagged release — under which the present Perspective and its supporting null-result case study are themselves produced.

1 The 2026 landscape

In 2026, two peer-reviewed *Nature* papers placed autonomous agentic-LLM research at the centre of scientific publishing. Google DeepMind's Co-Scientist¹ generated, critiqued and refined biomedical hypotheses, with acute-myeloid-leukaemia drug-repurposing candidates validated *in vitro*. Sakana AI's end-to-end pipeline² produced a manuscript that passed first-round workshop peer review under blind review; the authors' Limitations enumerate hallucinated citations and naive ideas. In parallel, a clinician with no large-lab affiliation can in 2026 assemble a workflow that performs every Co-Scientist task on commodity tooling — one domain prompt, a handful of orchestration prompts, an open-source LLM CLI — and produce a manuscript matched to a target journal's specifications. This Perspective and its supporting hepatocellular-carcinoma case study are the existence proof.

2 The disclosure gap

The biomedical-journal AI policy in current public form across the Lancet group, NEJM family and JAMA Network permits “readability and language improvements only”³ and prohibits using AI to generate scientific arguments, draft methodology, write literature reviews, or create new content. Read literally, the policy classifies almost every action a 2026 agentic workflow performs as prohibited. Read sympathetically, it protects against three real harms with rules calibrated for a 2023 generation of LLM use. *Authorship inflation*: listing an LLM as a co-author short-circuits the ICMJE accountability requirement; current policy correctly prohibits this and the Disclosure 2.0 proposal does not contest it. *Scientific hallucination*: fabricated citations, invented effect sizes and plausible-but-incorrect claims survive single-sentence acknowledgement; the Sakana group's authors enumerate this failure mode in their *Nature* paper's Limitations². *Undocumented intervention*: the reader of a 2026 methods section cannot tell which sentences were AI-drafted, which prompts produced them, or whether the cited literature actually exists. The policy's intent is sound; the mechanism is mis-calibrated. Prohibition without audit incentivises camouflage rather than catching the harm it fears. The reform is not stricter prohibition but more granular disclosure: make the activity visible, then audit it.

3 Disclosure 2.0

We propose that the published unit for AI-assisted biomedical research become a six-item manifest at a tagged release. (1) *The prompts*: every orchestration prompt committed verbatim, in order, attributed to the operator, closed for edits after first push. This is the central move; a prompt is what a methods paragraph used to be. (2) *Model identifier and snapshot*: vendor, model name, version, snapshot date, sampling parameters, and the list of tools the model was authorised to invoke. The combination is the reproducibility unit. (3) *The commit history*: a single repository whose git log is the chronological narrative, with force-pushes, rebases and history-rewriting forbidden. Mistakes are corrected by new commits whose messages name the prior mistake. (4) *Reviewer-subagent transcripts*: verbatim per round, including comments later overruled. (5) *The audit log*: a sealed subagent that re-executes the analysis from a clean clone, verifies every citation against Crossref or PubMed, and re-derives the headline statistics. Partial discharge of any one does not satisfy item 5. (6) *The tagged release*: a semantic version bundling the compiled manuscript, source, code, transcripts, audit log, and a Zenodo DOI. A minimum viable manifest (items 1, 2, 3, 6) covers the consequential audit surfaces at near-zero infrastructure cost and is the recommended on-ramp for clinician-investigators without git fluency or institutional compute. A vigorous human-in-the-loop school exemplified by the Academic Research Skills toolkit⁴ argues for AI under operator supervision rather than autonomous use; the two regimes are different equilibria with different cost-of-error profiles. Disclosure 2.0 is the orthogonal layer underneath both: an ARS-style copilot user emits the same six-item manifest that an autonomous-pipeline user does. Figure 1 is the artefact ledger of the present Perspective.

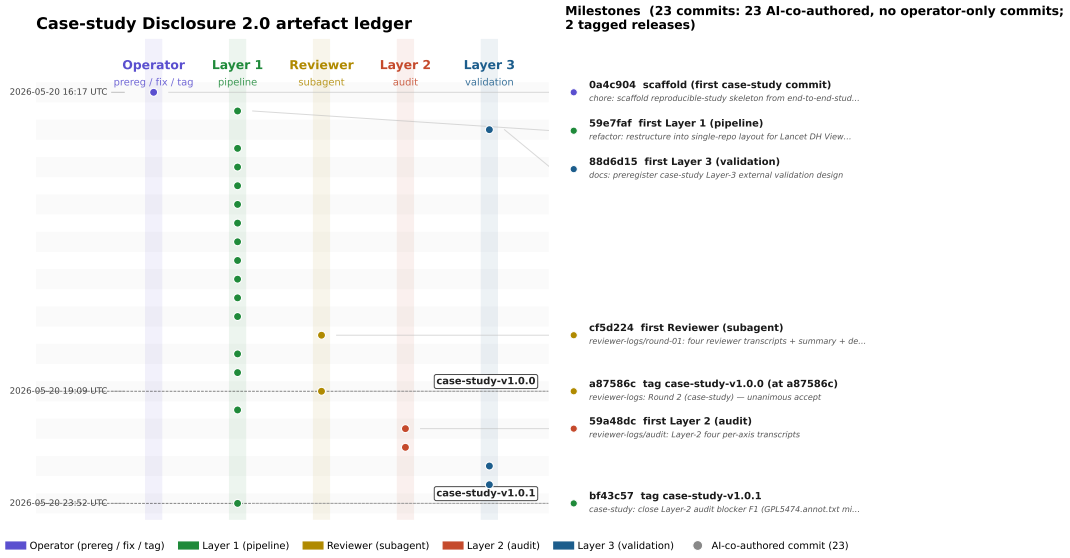


Figure 1: Disclosure 2.0 artefact ledger for the supporting hepatocellular-carcinoma case study only (the Viewpoint manuscript’s own commits and tags are deliberately excluded). 23 case-study commits across five swimlanes (Operator preregistration / fixes / tags, Layer 1 pipeline, Reviewer subagents, Layer 2 audit, Layer 3 validation); two tagged releases (**case-study-v1.0.0**, **case-study-v1.0.1**). Marker shape encodes commit authorship (circle = AI-co-authored, square = operator-only); colour encodes swimlane. The right panel lists milestone commits (first commit in each layer, tagged releases, HEAD). The ledger is regenerated from `git log` on each release tag.

4 The supporting null

The supporting case study runs Disclosure 2.0 on a hepatocellular-carcinoma overall-survival stratification problem outside the operator’s clinical sub-specialty (haematology and medical oncology). The cross-specialty choice is deliberate: TCGA-LIHC is MIT-redistributable whereas the operator’s hematology raw data is behind institutional firewalls, and the cross-domain mismatch is itself the

sharper test of whether the workflow generalises. Layer 1, a single autonomous Claude Code session, produced a 50-gene risk score (HCC-TRS) with development-cohort paired-optimism ΔC -index = 0.063 (95 % CI [0.003, 0.113], $n = 366$, 130 events). Layer 2 (a sealed audit subagent) filed twelve substantive findings: one blocker (a required GPL annotation file not downloaded by the data-prep script, so the external-cohort branch could not be regenerated from a clean clone), two high (an abstract figure quoting the legacy estimator rather than the paired-optimism estimator the Methods specify; a stratified-vs-stage-adjusted hazard ratio discrepancy of 7.34 vs 6.66), and nine medium and low (a fabricated citation DOI to an unrelated guideline; a Jaccard-stability claim disagreeing with the saved JSON; uncited bibliography entries). None of these were caught by the within-Layer-1 reviewer rounds, which closed at unanimous accept. Layer 3 (the operator’s preregistered⁵ external validation against held-out GSE14520 and GSE76427, pooled $n = 333$, 107 events) returned $\Delta C = 0.006$, 95 % CI [−0.008, 0.030] — primary hypothesis *not rejected*. Cohort-specific: $\Delta C = 0.016$ [−0.010, 0.047] (GSE14520) and $\Delta C = 0.026$ [−0.020, 0.141] (GSE76427). The seven preregistered secondaries (calibration at 1/3/5 y, decision-curve net benefit, IDI) are consistent with the null: calibration drifts from 1.1 at 1 y to 1.9 at 5 y, IDI is ≈ 0 . The development-cohort positive does not transport. The null is the headline; the manifest preserves it together with the twelve audit findings as evidence the regime makes its own failure modes visible. A case study that returns null *and* has twelve audit findings is the most rigorous test the disclosure standard can face. Two limits we acknowledge rather than deflect: Disclosure 2.0 does not claim it eliminates every failure mode — it claims only that it makes them findable, and findability still requires someone to look; and a single case study is not evidence the regime works at scale.

5 Recommendation

NEJM AI and the wider biomedical-journal community should pilot Disclosure 2.0 as an alternative submission track for AI-deep manuscripts. Implementation does not require Editorial Manager feature development: in the proposed 90-day pilot, submitting authors upload a Disclosure 2.0 PDF manifest alongside the cover letter, the editorial assistant verifies the linked Zenodo DOI against a JSON schema (a reference schema accompanies this Perspective in the linked repository at `docs/disclosure2-schema.json`), and the editorial office reports back on whether the manifest changes triage time or post-acceptance copy-edit burden. Five to ten Lancet DH or NEJM AI submissions per quarter are a sufficient sample. The pilot’s outcome metric is whether the manifest is actually *consulted* by reviewers and editorial staff, not merely published: a manifest no one reads has no disciplinary effect on the failure modes it is designed to catch. A complementary recommendation: where Layer 2 is the audit subagent, the manifest’s `models` array supports cross-model audit (Layer 1 on Anthropic, Layer 2 on Google Gemini or OpenAI GPT) and editorial offices should treat same-vendor audits as the single-rater-study equivalent. The Lancet group’s current bilateral prohibition on reviewer-side AI use should be lifted and replaced with the same disclosure regime; symmetric disclosure is the only stable equilibrium. The Perspective the reviewer is reading was assembled under exactly that standard; every artefact the manifest would require is publicly archived at <https://github.com/htlin222/end-to-end> under MIT licence at tag `viewpoint-nejmai-v1.0.0`, with a Zenodo DOI minted on tag push. The medium is the message; whatever the editorial decision, the artefact stands.

Declarations

Conflicts of interest. The author declares no competing interests. **Funding.** None. **Role of the funding source.** No funder; full repository access and final submission decision rest with the author. **AI usage.** Manuscript drafted with Anthropic Claude (Opus 4.7, 1M-context variant) via the Claude Code CLI; per-step disclosure log at `docs/ai-usage-disclosure.md` in the linked repository. No AI tool is listed as an author; no AI tool generated figures. **Data and code availability.** All artefacts at <https://github.com/htlin222/end-to-end> (MIT licence; tag `viewpoint-nejmai-v1.0.0`).

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